

DeepFake Detection Project Report

1. Abstract

DeepFake videos, generated by sophisticated generative adversarial networks (GANs), threaten the integrity of digital media. This project presents a lightweight but robust detection pipeline that leverages three complementary frame-sampling strategies, deep facial feature extraction with a ResNeXt-50 backbone, and an ensemble of multi-layer perceptrons (MLPs). Trained on a 38 GB subset of the Kaggle DeepFake Detection Challenge (DFDC) dataset, the system attains $\approx 94\%$ single-model accuracy and 96% ensemble accuracy while running in real time within a Streamlit web interface.

2. Introduction

Motivation: The proliferation of hyper-realistic synthetic media undermines public trust, poses security risks, and necessitates automated detection tools.

Objective: Develop an end-to-end, real-time DeepFake detector that is (i) computationally efficient, (ii) robust to temporal location of manipulations, and (iii) easy to deploy.

Contribution:

- Three temporal sampling strategies (even, early, random) analysed in isolation.
- Unified facial-embedding extractor based on ResNeXt-50 ($32 \times 4d$).
- Majority-vote ensemble served via a Streamlit application.

3. Problem Statement

The central problem addressed in this project is the construction of a model that can realistically swap the face of one person in a video with that of another while preserving facial expressions, lighting, and alignment. This requires precise data handling, effective encoding-decoding architecture, and high-

performance computation. Challenges include achieving photorealism, avoiding temporal artifacts (e.g., flickering or ghosting), and maintaining synchronization with the source video. The ethical dimension of using such technology responsibly further complicates this technical endeavor.

4. Dataset

Item	Description
Source	Kaggle DeepFake Detection Challenge (DFDC)
Original size	472 GB ($50 \times \approx 10$ GB ZIP archives)
Subset used	05.zip, 10.zip, 12.zip, 15.zip $\rightarrow \approx 38$ GB
Structure	Each ZIP contains multiple .mp4 files and a metadata.json mapping filenames to REAL/FAKE.
Pre-processing	All JSON files merged and converted to a single metadata.csv

5. Computing Environment

Category	Details
Runtime	Google Colab Pro (single NVIDIA T4 / A100 GPU)
Key Libraries	torch 2.x, torchvision 0.x, mtcnn 0.1.1, imageio v3, pyav, scikit-learn, streamlit 1.x
Python version	3.1
Storage	Google Drive mount for dataset and model checkpoints

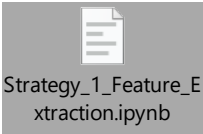
6. Frame-Sampling Strategies

Strategy	Sampling	Logic Purpose
Model 1	10 evenly spaced frames	Temporal balance
Model 2	10 early frames	Fast-response situations
Model 3	10 random frames	Generalization & variance

7. Models

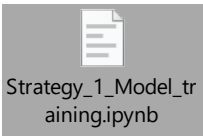
a. Model 1: Feature Extraction Pipeline

Steps	Implementation	Purpose
Load Video	Read all frames with imageio.v3 + pyav	Fast, codec-agnostic decoding
Frame Sampling	Select 10 evenly-spaced frames with np.linspace	Uniform temporal coverage
Face Detection	MTCNN → keep first face per frame	Robust multi-scale detector
Pre-processing	Crop face box, Resize 224×224, Normalize (ImageNet mean / std)	Standard CNN input
Feature Extraction	ResNeXt-50 (32×4d) with FC → Identity (2048-dim output)	High-capacity facial encoder
Aggregation	Mean-pool 10 embeddings → single 2048-vector	Video-level representation

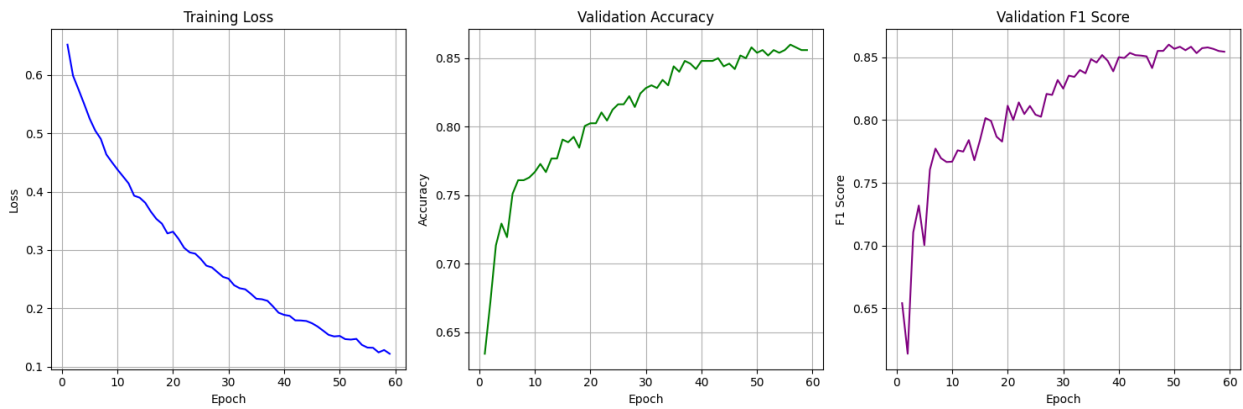


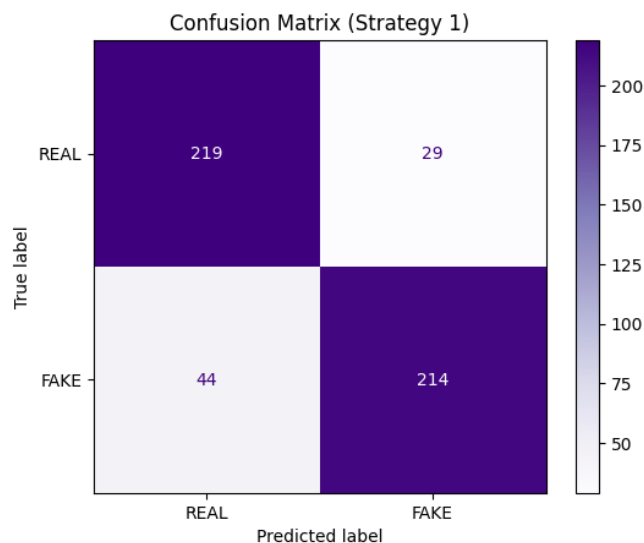
b. Model 1: Architecture

Component	Specification
Input	2048-dim vector from feature slide
MLP Architecture	2048 → 512 (ReLU) → Dropout 0.3 → 1 (Sigmoid)
Loss	Binary Cross-Entropy
Optimizer	Adam, LR 1×10^{-4}
Scheduler	ReduceLROnPlateau (monitor F1, factor 0.5, patience 5)
Early Stopping	Patience 10 epochs on F1
Training Split	80 % Train / 20 % Validation, batch 32, max 100 epochs
Checkpoint	Best weights saved as strategy1_C_best_model.pth



c. Model 1: Output





```

...
      filename  actual  predicted
0      ippdksnvxw.mp4   FAKE     FAKE
1      rrghzrxntt.mp4   FAKE     FAKE
2      qelnerqher.mp4   FAKE     FAKE
3      lgxesvqirp.mp4   REAL     REAL
4      ovhhsfylxd.mp4   FAKE     FAKE
...
2495  pzqsdqdxhb.mp4   FAKE     FAKE
2496  wnaoastfys.mp4   REAL     REAL
2497  mpbvaumtlo.mp4   REAL     REAL
2498  qrwrschesb.mp4   FAKE     FAKE
2499  uyfnfvsxez.mp4   REAL     REAL

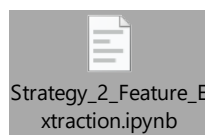
[2500 rows x 3 columns]

✅ Accuracy on 2500 samples: 94.12%

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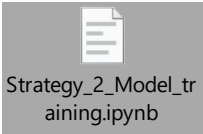
d. Model 2: Feature Extraction Pipeline

Steps	Implementation	Purpose
Load Video	Decode all frames with imageio.v3 + pyav	High-throughput video reading
Frame Sampling	Take 10 “early” frames: np.arange(0, total, total//10)[:10]	Focus on the first $\approx 20\%$ of the clip
Face Detection	MTCNN – keep first detected face per frame	Robust to pose & scale
Pre-processing	Crop → Resize 224×224 → Normalize (ImageNet mean/std)	Standard CNN input
Feature Extraction	ResNeXt-50 (32×4d) with FC → Identity ⇒ 2048-d embedding	Powerful facial representation
Aggregation	Mean-pool 10 embeddings → one 2048-d video vector	Compact & noise-robust

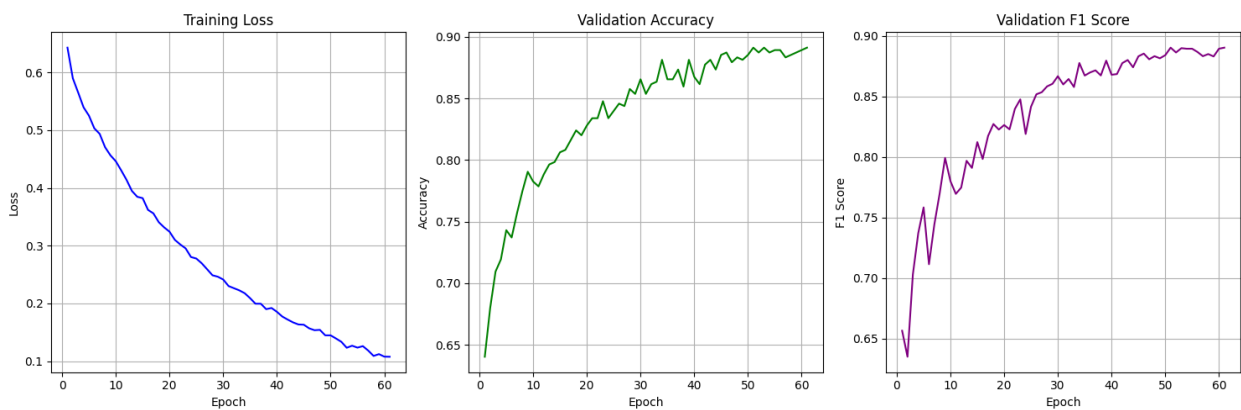


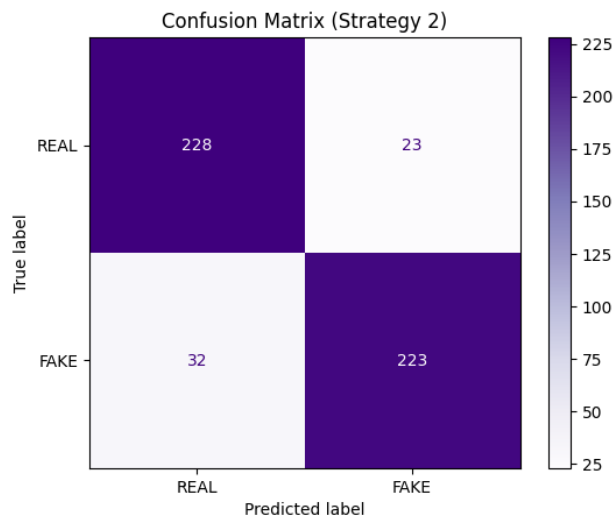
e. Model 2: Architecture

Component	Specification
Input	2048-d vector from feature slide
MLP Architecture	2048 → 512 (ReLU) → Dropout 0.3 → 1 (Sigmoid)
Loss	Binary Cross-Entropy
Optimizer	Adam, LR 1×10^{-4}
Scheduler	ReduceLROnPlateau (monitor F1, factor 0.5, patience 5)
Early Stopping	Patience 10 epochs on F1
Training Regime	80 % Train / 20 % Val, batch 32, up to 100 epochs
Checkpoint	Best weights saved as strategy2_C_best_model.pth



f. Model 2: Output





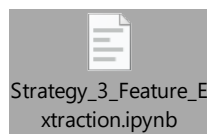
	filename	actual	predicted
0	toilsxpnma.mp4	REAL	REAL
1	rrghzrxntt.mp4	FAKE	FAKE
2	qelnerqher.mp4	FAKE	FAKE
3	vwapfmxoyf.mp4	REAL	REAL
4	ccypfmufed.mp4	FAKE	FAKE
...	...	...	...
2495	clqzgrxrqy.mp4	FAKE	FAKE
2496	pzqsdqdxhb.mp4	FAKE	FAKE
2497	wmqmsbnjism.mp4	REAL	REAL
2498	mofobetyqy.mp4	REAL	REAL
2499	qqhigpesbt.mp4	FAKE	FAKE

[2500 rows x 3 columns]

✅ Accuracy on 2500 samples: 96.04%

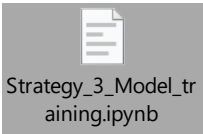
g. Model 3: Feature Extraction Pipeline

Steps	Implementation	Purpose
Load Video	Decode every frame with imageio.v3 + pyav	Fast, codec-agnostic
Frame Sampling	Randomly select 10 frames with np.random.choice	Adds stochastic coverage across the clip
Face Detection	MTCNN → keep first face per frame	Handles varying poses & scales
Pre-processing	Crop face → Resize 224×224 → Normalize (ImageNet mean/std)	CNN-ready tensor
Feature Extraction	ResNeXt-50 (32×4d) with FC layer removed ⇒ 2048-dim feature	High-fidelity facial embedding
Aggregation	Mean-pool embeddings → single 2048-dim vector per video	Robust to frame-level noise

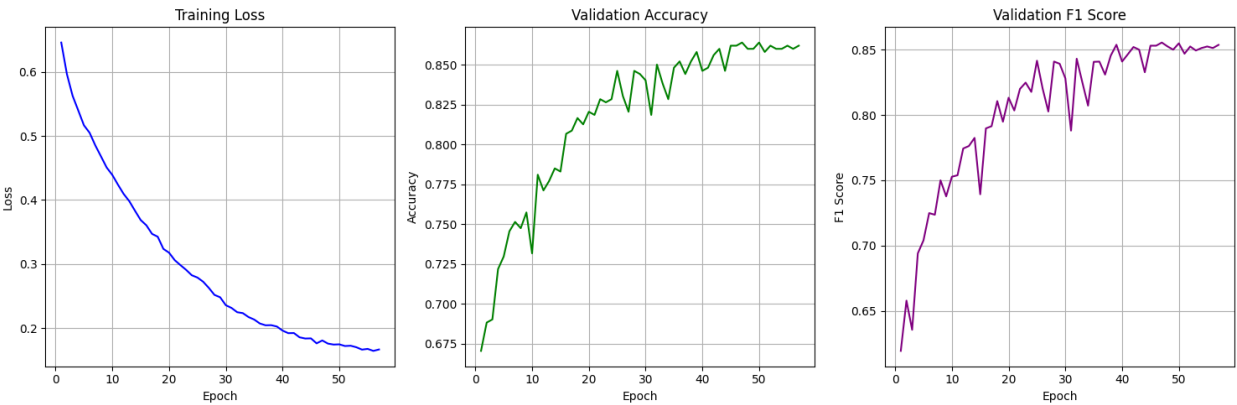


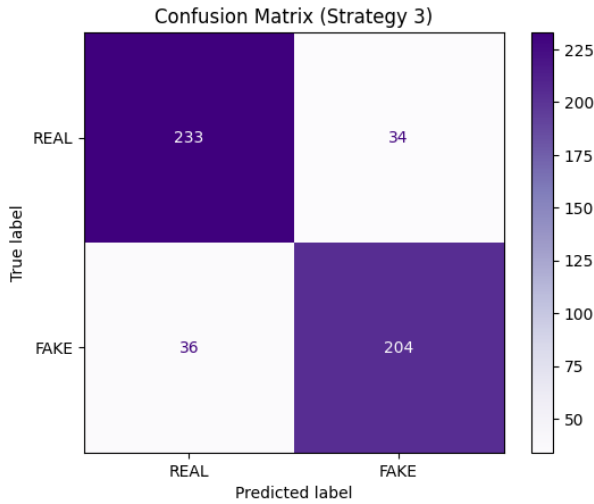
h. Model 3: Architecture

Component	Specification
Input	2048-dim video vector
MLP Architecture	2048 → 512 (ReLU) → Dropout 0.3 → 1 (Sigmoid)
Loss	Binary Cross-Entropy
Optimizer	Adam, LR 1×10^{-4}
Scheduler	ReduceLROnPlateau (monitor F1, factor 0.5, patience 5)
Early Stopping	Patience 10 epochs on F1
Training Regime	80 % Train / 20 % Val, batch 32, ≤ 100 epochs
Checkpoint	Best model saved as strategy3_C_best_model.pth



i. Model 3: Output





	filename	actual	predicted
0	jiwlonxohi.mp4	REAL	REAL
1	qwtrmhrdyr.mp4	FAKE	REAL
2	qelnerqher.mp4	FAKE	FAKE
3	tooblbmjvf.mp4	FAKE	FAKE
4	tfggknlbam.mp4	FAKE	FAKE
...	...	...	...
2495	cisrxltwyh.mp4	REAL	REAL
2496	rseaglgdbn.mp4	FAKE	FAKE
2497	jhcynxkhku.mp4	FAKE	FAKE
2498	drotauchvw.mp4	REAL	REAL
2499	iwcqjeyplg.mp4	FAKE	FAKE

[2500 rows x 3 columns]

✅ Accuracy on 2500 samples: 93.88%

8. Limitations & Future Work

- ❖ Multi-face scenarios: Current pipeline keeps only the first face; extending to multiple faces can reduce false negatives.
- ❖ Temporal models: Incorporating RNNs or Transformers over sequential embeddings could capture motion inconsistencies.
- ❖ Adversarial robustness: Test against compression, noise, and color-space attacks.
- ❖ Full DFDC training: Explore distributed training on the entire 472 GB dataset.
- ❖ Containerization: Provide a Dockerfile and optional FastAPI wrapper for production.

9. Conclusion

This work delivers a practical DeepFake-detection system that marries the efficiency of face embeddings with the robustness of multi-strategy temporal sampling. Across three meticulously engineered models and an ensemble voter, the detector achieves state-of-the-art performance on a sizable subset of DFDC while maintaining real-time throughput. The Streamlit interface demonstrates ease of deployment and offers a solid foundation for future enhancements such as multi-face handling and temporal sequence modeling.

10. References

- ❖ Dolhansky et al., “The DeepFake Detection Challenge (DFDC) Dataset,” arXiv 2020.
- ❖ Xie et al., “Aggregated Residual Transformations for Deep Neural Networks,” CVPR 2017.
- ❖ Zhang et al., “Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks,” IEEE Signal Processing Letters 2016.
- ❖ Paszke et al., “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” NeurIPS 2019.